

Reviewer Pack — Minimal Root Separation of Quaternion Algebras vs. AC⁰ Parit...

Entry 9681c9c31039 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Root Separation of Quaternion Algebras vs. AC⁰ Parity Threshold

Entry ID: 9681c9c31039 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 12:16:31 UTC

1. Conjecture

Field A (mathematical branch): Quaternion algebra theory **Field B** (complexity object): Complexity Theory: AC⁰ Parity Circuit Complexity

Statement:

[‘For any Boolean function f computable by an AC⁰ parity circuit of size n , the minimal norm separating two non-trivial quaternionic embeddings of the polynomial defining f is at least $\Omega(n^c)$ for some constant $c > 0$.’, ‘The minimal norm is defined as the infimum of norms of all isometric embeddings of the polynomial into a quaternion algebra over the complex numbers.’, ‘An AC⁰ parity circuit computes a function with parity gates only, where a parity gate is a XOR operation followed by a NOT gate.’]

Rationale (proposer’s reasoning):

[‘Quaternion algebras provide a non-commutative extension of the complex numbers that can be used to represent certain types of polynomials. It has been observed that non-commutative structures can lead to increased complexity in computations, potentially explaining the hardness of problems like PARITY.’, ‘If the conjecture is true, it would imply that the AC⁰ parity threshold for PARITY is related to the structure of quaternion algebras, offering a new perspective on circuit complexity.’, ‘This could potentially lead to new algorithms or lower bounds for AC⁰ parity circuits.’]

Taxonomy category: AC⁰_PARITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): e87cfec12fe72001

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, across 30 random seeds, at least 80% of generated AC⁰ parity circuits with size $n \leq 40$ have a minimal norm separating two non-trivial quaternionic embeddings of their defining polynomial that is at least $\Omega(n^c)$ for some constant $c > 0$. The conjecture is falsified if any seed produces a minimal norm less than $\Omega(n^c)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	1.00	UNCERTAIN	SAFE
ALGEBRIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal norm separation" AND quaternion algebra theory AND AC⁰ parity circuit" - "AC⁰ parity threshold" AND quaternionic embedding AND polynomial" - "isometric embeddings" IN quaternion algebra theory AND AC⁰ parity circuits

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_random_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def ac0_parity_circuit_size(f):
        n = len(f)
        if n == 1:
            return 1
        else:
            return 1 + max(ac0_parity_circuit_size(f[:n//2]), ac0_parity_circuit_size(f[n//2:]))

    def polynomial_from_function(f):
        n = len(f)
        x = [i for i in range(n)]
        poly = f[0]
        for i in range(1, n):
            term = f[i] * math.prod(x[j] if j < i else -x[j] for j in range(i))
            poly += term
        return poly

    def quaternion_norm(poly):
        norm = 0
        for coeff in poly:
            norm += abs(coeff)**2
        return math.sqrt(norm)
```

```

n_values = [5, 10, 15, 20, 30, 40]
instances_tested = 0
conjecture_holds = True
counterexample = ""

for n in n_values:
    for _ in range(5): # Ensure at least 30 instances per seed
        f = generate_random_boolean_function(n)
        circuit_size = ac0_parity_circuit_size(f)
        if circuit_size > n:
            continue
        poly = polynomial_from_function(f)
        norm = quaternion_norm(poly)
        if norm < n**0.5: # Simplified bound for demonstration
            conjecture_holds = False
            counterexample = f"n={n}, f={f}, norm={norm}"
            break
    instances_tested += len(n_values) * 5

return {
    "metric_name": "quaternion_norm",
    "metric_value": n**0.5, # Simplified bound for demonstration
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample='{result['counterexample']}' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it did not complete within the allotted time frame. | next: NONE

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14681
2	preregistration	ollama_remote	glm4:latest	0	0	9334
3	novelty	ollama_remote	glm4:latest	0	0	8273
4	novelty	ollama_remote	glm4:latest	0	0	8446
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14775
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10474
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7572
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9170
9	judge	ollama_remote	glm4:latest	0	0	19823

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 102548 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/9681c9c31039.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/9681c9c31039.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/9681c9c31039.tar.gz* (if generated)